

A COUNTEREXAMPLE TO A CONJECTURE OF THAKUR ON CARLITZ–WIEFERICH PRIMES

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ABSTRACT. Let $\mathbf{A} = \mathbf{F}_q[T]$ with q a power of an odd prime p , let $[n] = T^{q^n} - T$, and let ρ be the Carlitz module. A monic prime P of $\mathbf{A}$ is a *c-Wieferich prime* (to base 1) if $\rho_P(1) \equiv 1 \pmod{P^2}$. Thakur suggested in 2015, on the basis of limited data and of proofs in degrees 2 and 3, that in odd characteristic every *c-Wieferich prime* has degree divisible by p ; the question was restated as open in 2024, and Bamunoba and Bergström, after extensive computations, expressed the belief that the statement holds in odd characteristic. We show that it is false: the polynomial

$$P(T) = T^5 + (11 + 17c + 9c^2)T^4 + (3 + 7c + 18c^2)T^3 + (2 + 5c + 6c^2)T^2 + (3 + 3c + 11c^2)T + (6 + 17c + 5c^2)$$

over $\mathbf{F}_{19^3} = \mathbf{F}_{19}[c]$, $c^3 = 8c^2 + 4c + 11$, is an irreducible *c-Wieferich prime* of degree 5, and $19 \nmid 5$. We further give a closed form for the resulting common factor: the polynomial $G = \mu(T^q - T)$, with $\mu(X) = X^5 + 5X^3 + 3X^2 - 4X - 9 \in \mathbf{F}_{19}[X]$ and $q = 19^3$, is squarefree of degree $5 \cdot 19^3$ and divides both $[5]$ and Thakur's polynomial M_5 ; we conjecture that in fact $G = \gcd([5], M_5)$, and we verify this for the part of low degree over the prime field. Degree 5 is the least possible degree of such a counterexample, and exhaustive computations show that no counterexample exists over the prime fields $\mathbf{F}_p$ in a substantial range of degrees and characteristics. The proof that degree 5 is minimal, and the method by which the example was found, appear in a companion paper.

1. INTRODUCTION

Let p be an odd prime, q a power of p , and $\mathbf{A} = \mathbf{F}_q[T]$. Write $[n] = T^{q^n} - T$ for $n \geq 1$. The Carlitz module is the $\mathbf{F}_q$ -algebra homomorphism $\rho: \mathbf{A} \rightarrow \mathbf{A}\{\tau\}$ into the twisted polynomial ring ($\tau x = x^q \tau$) determined by $\rho_T = T + \tau$; for $a \in \mathbf{A}$ we write $\rho_a(x)$ for the associated $\mathbf{F}_q$ -linear polynomial. Following Thakur [6] and Bamunoba–Bergström [2], a monic prime $P \in \mathbf{A}$ is called a *c-Wieferich prime* (to base 1) if

$$(1) \quad \rho_P(1) \equiv 1 \pmod{P^2}.$$

(The congruence $\rho_P(1) \equiv 1 \pmod{P}$ holds for every prime P , this being the Carlitz analogue of Fermat's little theorem; (1) is the analogue of the classical Wieferich condition $2^{p-1} \equiv 1 \pmod{p^2}$.)

In [6], Thakur introduced the quantities

$$M_d = \sum_{k=0}^{d-1} (-1)^k [d-1][d-2] \cdots [d-k] \quad (\text{the } k=0 \text{ term being } 1),$$

proved that no *c-Wieferich prime* of degree 2 or 3 exists in odd characteristic, and suggested, from this and limited numerical data, that for $p > 2$ a nontrivial common factor of $[d]$ and M_d should force $p \mid d$ — equivalently, that every *c-Wieferich prime* of $\mathbf{A}$ has degree divisible by p (see Lemma 2.1 below for the equivalence). The suggestion fails in characteristic 2, by examples of Mauduit (see [6, 2]); Bamunoba and Bergström carried out extensive computations and stated that they believe it to be true in odd characteristic [2]. The question was restated as open by Thakur in 2024 [7].

The purpose of this note is to show that the statement is false in odd characteristic.

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Theorem 1.1. *Let $q = 19^3$ and $\mathbf{F}_q = \mathbf{F}_{19}[c]$ with $c^3 = 8c^2 + 4c + 11$. The polynomial*

$$P(T) = T^5 + (11 + 17c + 9c^2)T^4 + (3 + 7c + 18c^2)T^3 + (2 + 5c + 6c^2)T^2 + (3 + 3c + 11c^2)T + (6 + 17c + 5c^2)$$

is an irreducible c -Wieferich prime of degree 5 in $\mathbf{F}_{19^3}[T]$. Consequently $\gcd([5], M_5) \neq 1$ in $\mathbf{F}_{19^3}[T]$ although $19 \nmid 5$. Moreover P is not fixed by any nontrivial additive translation $T \mapsto T + \gamma$.

The last assertion is automatic (Lemma 2.5) and is recorded because of its significance: the *fixed* c -Wieferich primes, i.e. those invariant under a translation, necessarily have degree divisible by p , and the computations of [2] produced only fixed examples in odd characteristic; Theorem 1.1 therefore also answers negatively the stronger expectation, formulated in [2], that no non-fixed c -Wieferich prime exists in odd characteristic.

The example does not come from a search: the space of candidates is far beyond exhaustion (about $3 \cdot 10^{18}$ monic quintics over $\mathbf{F}_{19^3}$, or $19^{12} \approx 2.2 \cdot 10^{15}$ candidates in the parametrization of [2, Thm. 3.4]). It was produced by a systematic method which reduces the existence of a c -Wieferich prime of prescribed degree in a prescribed odd characteristic to a finite, effective computation, and which explains both the non-existence results in low degree and the sporadic failure exhibited here. The method, together with the proof of Theorem 5.1 below, is presented in the companion paper [4], where the exactness assertion (Conjecture 4.2) is also reduced to a single explicit question and verified in the range accessible to exact computation.

Beyond the counterexample itself, we prove here a closed form for the resulting common factor (Theorem 4.1: it is $\mu(T^q - T)$ for an explicit quintic μ with coefficients in the prime field), we discuss the minimality of the degree (degrees 1, 2 and 3 are excluded by [6] (degree 1 trivially; see also [1]), and degree 4 by Theorem 5.1), and we report exhaustive computations showing that no counterexample exists over the prime fields $\mathbf{F}_p$ themselves in a substantial range (Proposition 5.2).

All computations in this paper are exact (no floating point), were carried out in standard public computer-algebra systems, and were reproduced on independent engines; see Section 6 and the Appendix, which contains complete verification code. A statement on the methodology of this work, including the role of AI assistance, is given in Section 6.

2. PRELIMINARIES

We collect standard facts and elementary lemmas. Throughout, q is a power of the odd prime p , and θ denotes a root of the monic prime $P \in \mathbf{A} = \mathbf{F}_q[T]$ of degree d , so that $\mathbf{F}_q(\theta) = \mathbf{F}_{q^d}$.

Following [2], define $F_0 = 1$ and $F_i = (-1)^i + [i]F_{i-1}$ for $i \geq 1$. An immediate induction gives $F_{d-1} = (-1)^{d-1}M_d$, so $P \mid M_d$ if and only if $P \mid F_{d-1}$. Evaluating at θ and writing

$$y_j = [j](\theta) = \theta^{q^j} - \theta \quad (1 \leq j \leq d-1),$$

one obtains the nested form

$$(2) \quad (-1)^{d-1}F_{d-1}(\theta) = M_d(\theta) = 1 - y_{d-1}(1 - y_{d-2}(\cdots(1 - y_1)\cdots)).$$

Lemma 2.1 ([2, Prop. 1.2]). *A monic prime P of degree d is a c -Wieferich prime if and only if $P \mid F_{d-1}$, equivalently $P \mid M_d$, equivalently $M_d(\theta) = 0$.*

Lemma 2.2. $[n] = T^{q^n} - T$ *is the product of all monic primes of $\mathbf{A}$ of degree dividing n ; in particular $[n]$ is squarefree, and every irreducible polynomial of degree n divides $[n]$. (See, e.g., [5, Ch. 2].)*

Lemma 2.3. $\gcd([d], M_d)$ *has no factor of degree 1. Indeed, for $a \in \mathbf{F}_q$ one has $[i](a) = a^{q^i} - a = 0$ for all $i \geq 1$, whence $M_d(a) = 1 \neq 0$.*

Lemma 2.4 (Translation invariance). *Let P be a c -Wieferich prime of degree d with root θ , and let $a \in \mathbf{F}_q$. Then $P(T - a)$, the minimal polynomial of $\theta + a$, is again a c -Wieferich prime of degree d .*

Proof. $[j](\theta + a) = (\theta + a)^{q^j} - (\theta + a) = \theta^{q^j} - \theta = y_j$ for every j , since $a^{q^j} = a$. Hence $M_d(\theta + a) = M_d(\theta) = 0$, and Lemma 2.1 applies to the prime $P(T - a)$. $\square$

Lemma 2.5. *If $p \nmid d$, then $P(T + \gamma) \neq P(T)$ for every $\gamma \in \mathbf{F}_q^\times$; consequently the q translates $P(T - a)$, $a \in \mathbf{F}_q$, are pairwise distinct.*

Proof. Comparing the coefficients of T^{d-1} in $P(T + \gamma)$ and $P(T)$ gives $d\gamma = 0$, hence $\gamma = 0$ when $p \nmid d$. $\square$

Lemma 2.6 (Norm form of a translation class). *Let P be monic of degree d with roots $\theta_1, \dots, \theta_d$ (in an algebraic closure), and set $R_P(X) = \prod_{j=1}^d (X - (\theta_j^q - \theta_j)) \in \mathbf{F}_q[X]$, which equals the characteristic polynomial of multiplication by $\theta^q - \theta$ on $\mathbf{A}/(P)$. Then*

$$\prod_{a \in \mathbf{F}_q} P(T - a) = R_P(T^q - T).$$

Proof. Using $\prod_{a \in \mathbf{F}_q} (Y - a) = Y^q - Y$ with $Y = T - \theta_j$,

$$\prod_{a \in \mathbf{F}_q} P(T - a) = \prod_j \prod_a (T - a - \theta_j) = \prod_j ((T - \theta_j)^q - (T - \theta_j)) = \prod_j (T^q - T - (\theta_j^q - \theta_j)).$$

$\square$

Finally we record the parametrization underlying the exhaustive computations of Section 5; it is essentially [2, Thms. 3.3–3.4], formulated additively.

Lemma 2.7. *Let $d \geq 2$. There exists a c -Wieferich prime of degree d in $\mathbf{F}_q[T]$ if and only if there exists $\eta \in \mathbf{F}_{q^d}$ with $\text{Tr}_{\mathbf{F}_{q^d}/\mathbf{F}_q}(\eta) = 0$, whose partial sums $s_j = \eta + \eta^q + \dots + \eta^{q^{j-1}}$ are nonzero for $1 \leq j \leq d-1$, and which satisfies $1 - s_{d-1}(1 - s_{d-2}(\dots(1 - s_1)\dots)) = 0$.*

Proof. If P is such a prime with root θ , take $\eta = \theta^q - \theta$; then $s_j = y_j$, the trace of η is $y_d = 0$, the s_j ($j < d$) are nonzero because θ has degree exactly d , and the displayed expression is $M_d(\theta) = 0$ by (2) and Lemma 2.1. Conversely, given such an η , the additive Hilbert 90 theorem provides $\theta \in \mathbf{F}_{q^d}$ with $\theta^q - \theta = \eta$; its partial sums are the s_j , so θ has degree exactly d , and its minimal polynomial is a c -Wieferich prime of degree d by (2) and Lemma 2.1. $\square$

3. THE COUNTEREXAMPLE

Proof of Theorem 1.1. The three assertions to be checked are finite exact computations: (i) P is irreducible over $\mathbf{F}_{193}$; (ii) $\rho_P(1) \equiv 1 \pmod{P^2}$, i.e. P is c -Wieferich by definition (1); equivalently (Lemma 2.1) $P \mid M_5$; (iii) $19 \nmid 5$ (clear), and P is non-fixed by Lemma 2.5.

For (ii) we verified both the definition (1) directly, by computing the coefficients of ρ_P through the Horner scheme $\rho_{Tf+a} = \rho_T \circ \rho_f + a$ with all coefficient arithmetic reduced modulo P^2 , and the criterion $P \mid M_5$, by computing the brackets $[i]$ modulo P through modular exponentiation. Complete code performing (i)–(ii) in a few seconds appears in the Appendix; the same checks were reproduced independently in Sage, PARI/GP, FLINT and the `galois` Python library; the implementations of the criteria were calibrated on the known c -Wieferich prime $T^6 + T^4 + T^3 + T^2 + 2T + 2$ of $\mathbf{F}_3[T]$ [2] (positive control) and on ordinary primes (negative controls). Since P is irreducible of degree 5, it divides [5] (Lemma 2.2), so $P \mid \gcd([5], M_5)$ and the latter is nontrivial. $\square$

Remark 3.1. As D. Thakur pointed out to the author, the existence of a c -Wieferich prime of degree 5 already follows from $\gcd([5], M_5) \neq 1$ together with Lemmas 2.2 and 2.3, without exhibiting P : any nontrivial common factor is a product of primes of degree 5 dividing M_5 , and each of these is c -Wieferich by Lemma 2.1. The explicit P is nevertheless recorded, both as a certificate and for its own interest.

4. A CLOSED FORM FOR THE COMMON FACTOR

Since the criterion of Lemma 2.1 is invariant under $\theta \mapsto \theta + a$ (Lemma 2.4), the counterexample comes with a whole translation class of $q = 19^3$ pairwise distinct c -Wieferich primes $P(T - a)$ (Lemma 2.5). Their product has a strikingly simple closed form.

Theorem 4.1. *Let $q = 19^3$, let P be as in Theorem 1.1, and set*

$$\mu(X) = X^5 + 5X^3 + 3X^2 - 4X - 9 \in \mathbf{F}_{19}[X], \quad G = \mu(T^q - T) \in \mathbf{F}_q[T].$$

Then:

- (a) $G = \prod_{a \in \mathbf{F}_q} P(T - a)$; in particular G is squarefree of degree $5 \cdot 19^3 = 34295$, a product of 6859 distinct c -Wieferich primes of degree 5;
- (b) G divides $[5]$ and G divides M_5 ; hence $G \mid \gcd([5], M_5)$.

Note that μ has coefficients in the prime field $\mathbf{F}_{19}$, not merely in $\mathbf{F}_{19^3}$.

Proof. (a) By Lemma 2.6, $\prod_a P(T - a) = R_P(T^q - T)$ where R_P is the characteristic polynomial of $\theta^q - \theta$ on $\mathbf{F}_q[T]/(P)$; a direct exact computation (three lines of Sage; see the Appendix) gives $R_P = \mu$. The factors $P(T - a)$ are irreducible of degree 5, pairwise distinct by Lemma 2.5, and c -Wieferich by Lemma 2.4.

(b) Each factor $P(T - a)$ divides $[5]$ (Lemma 2.2) and divides M_5 (Lemmas 2.4 and 2.1); since the factors are pairwise coprime, their product G divides both. $\square$

The checks of Theorem 4.1(b) were, in addition, performed directly at degree 34295: the congruences $T^{q^5} \equiv T \pmod{G}$ and $M_5 \equiv 0 \pmod{G}$, as well as $P \mid G$ and the squarefreeness of G , were verified independently in FLINT and in PARI/GP.

We conjecture that nothing else divides the gcd.

Conjecture 4.2. $\gcd([5], M_5) = G$ in $\mathbf{F}_{19^3}[T]$. Equivalently, the c -Wieferich primes of degree 5 in $\mathbf{F}_{19^3}[T]$ are exactly the 6859 translates $P(T - a)$, $a \in \mathbf{F}_{19^3}$.

By Lemmas 2.2 and 2.3, Conjecture 4.2 amounts to the statement that the translation class of P exhausts the degree-5 c -Wieferich primes of $\mathbf{F}_{19^3}[T]$. This is a completeness statement about a search space of size 19^{12} (Lemma 2.7). In the companion paper [4] the method of Section 5 reduces it to a single explicit question — the possible existence of a degree-5 c -Wieferich prime over $\mathbf{F}_{19^3}$ whose associated $\eta = \theta^q - \theta$ has degree 15 over $\mathbf{F}_{19}$ — and settles the part in which η has degree at most 5 over $\mathbf{F}_{19}$, which the class of P exhausts. The remaining case would require a gcd computation between polynomials of degree of order $3 \cdot 10^{11}$ and $2 \cdot 10^{15}$ over $\mathbf{F}_{19}$, or an enumeration of $\mathbf{F}_{19^{15}}$; both lie beyond exact computation, and the conjecture is left open.

5. MINIMALITY OF THE DEGREE, AND THE PRIME FIELDS

Degree 1 is excluded, in every odd characteristic, in [6] (the case of degree 1 being trivial; it also appears as [1, Cor. 4.5]). As Thakur observes, degree 1 is in fact immediate from Lemma 2.3: a linear prime would have to divide $\gcd([d], M_d)$, which has no linear factor since $M_d(a) = 1$ for every $a \in \mathbf{F}_q$. Degrees 2 and 3 are excluded, in every odd characteristic, in [6]. The next case is settled by the method of [4]:

Theorem 5.1 (proved in [4]). *There is no c -Wieferich prime of degree 4 in $\mathbf{F}_q[T]$, for any power q of any odd prime.*

Consequently 5 is the least possible degree of a counterexample in odd characteristic, and Theorem 1.1 is degree-minimal.

It is natural to ask (as D. Thakur did, in correspondence) whether a counterexample exists over a prime field $\mathbf{F}_p$ itself. Lemma 2.7 reduces this, for each pair (d, p) , to an exhaustive computation

over the p^{d-1} trace-zero elements of $\mathbf{F}_{p^d}$, which we carried out by exact vectorized arithmetic. Each cell of the following table is a definitive determination, not a partial search.

Proposition 5.2. *There is no c -Wieferich prime of degree d in $\mathbf{F}_p[T]$, for every pair (d, p) with $p \nmid d$ in the following list:*

d	characteristics p (exhaustively verified, $q = p$)
5	3, 7, 11, 13, 17, 19, 23, 29, 31, 37
6	5, 7, 11, 13, 17, 19, 23
7	3, 5, 11, 13
8	3, 5, 7
9	5
10	3
11	3, 5
13	3
14	3

For the conforming cell $(d, p) = (5, 5)$, the same scan finds precisely the known c -Wieferich prime $T^5 + 4T + 1$ of $\mathbf{F}_5[T]$ [2], and for $(d, p) \in \{(6, 3), (7, 7)\}$ it recovers the known primes $T^6 + T^4 + T^3 + T^2 + 2T + 2$ and $T^7 + 6T + 3$; these serve as positive controls of the implementation.

The reduction of Lemma 2.7 makes further prime-field cells accessible by the same exact computation, and the empty region above is being extended. The example of Theorem 1.1 lives over a cubic extension in an essential way, and nothing in the present data points either towards or against the existence of a prime-field example.

6. METHODOLOGY AND USE OF AI ASSISTANCE

This work was carried out by the author, an independent researcher with no formal mathematical training, in sustained collaboration with an AI assistant (Claude Opus 4.8, Anthropic), used through its standard paid consumer interface. The division of labour, stated here in the interest of full transparency and in line with editorial policies on AI-assisted work, was as follows. The author, who has no formal mathematical training, set the research direction and the criteria for the problems to be pursued, and contributed a structural, visual reading of the objects involved; the AI assistant proposed specific problems meeting those criteria, and supplied the mathematical domain knowledge, the formalization, the drafting, and the design and execution of all computations, under the author's direction. The strategy for each problem emerged from the dialogue between the two. Lacking the training to verify the mathematics directly, the author relied throughout on exact computational checks, on reproduction across independent systems, and, for the present result, on the independent verification kindly carried out by D. Thakur. All computations were performed in standard public computer-algebra systems (SageMath, PARI/GP, FLINT, and the `galois` Python library), in exact arithmetic, and every principal claim was reproduced on at least two mutually independent engines after calibration on known cases from the literature. Complete verification code for the main theorem is given in the Appendix, and all scripts are available from the author.

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APPENDIX A. VERIFICATION CODE

The following self-contained Sage code verifies Theorem 1.1 and the computation $R_P = \mu$ of Theorem 4.1(a) in a few seconds.

```

p = 19; q = p^3
R0.<x> = GF(p) []
F.<c> = GF(p^3, modulus = x^3 - 8*x^2 - 4*x - 11)
R.<T> = PolynomialRing(F)
P = (T^5 + (11+17*c+9*c^2)*T^4 + (3+7*c+18*c^2)*T^3
      + (2+5*c+6*c^2)*T^2 + (3+3*c+11*c^2)*T + (6+17*c+5*c^2))
print(P.is_irreducible())           # irreducibility
S = R.quotient(P); t = S.gen()
M = S(1)
for i in range(1, 5): M = 1 - (t^(q^i) - t)*M
print(M == 0, t^(q^5) - t == 0)    # P | M_5 and P | [5]
S2 = R.quotient(P^2); u = S2.gen()
rho = []                            # Carlitz rho_P by Horner
for cj in P.coefficients(sparse=False)[::-1]:
    new = [S2(0)]*(len(rho)+1)
    for i, a in enumerate(rho):
        new[i] += u*a
        new[i+1] += a.lift()(u^q)    # a -> a^{(q)}: F_q-coeffs are fixed
    new[0] += S2(cj)
    rho = new
D = (sum(rho) - 1).lift()
print(D % P == 0 and (D // P) % P == 0) # v_P(rho_P(1)-1) >= 2
mu = (t^q - t).matrix().charpoly('X')   # R_P of Lemma 2.6
print(mu)                                # X^5+5X^3+3X^2-4X-9

```

For Theorem 4.1(b) at full degree 34295, the congruences $T^{q^5} \equiv T \pmod{G}$ and $M_5 \equiv 0 \pmod{G}$ with $G = \mu(T^q - T)$ are a handful of square-and-multiply passes modulo G ; they were carried out in FLINT and in PARI/GP.

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